

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 13 · LECTURE NOTES

What a Photograph Is Now

Measurement, inference, and what you can claim.

MODULE OUTCOMES

Articulate where the boundary between photography and synthesis sits.

Explain provenance approaches and their limits.

Evaluate computational imaging claims critically.

State the obligations that follow for a practitioner.

Situate this course's methods against those questions.

1 Where the boundary sits

PHOTOGRAPHY WAS NEVER A PURE MEASUREMENT

Film had a characteristic response curve — non-linear, with a toe and a shoulder — a particular spectral sensitivity that did not match the eye, and a development process with substantial latitude in it.

Dodging and burning are as old as the darkroom. Ansel Adams's prints are heavily and deliberately interpreted — his zone system is a methodology for deciding what the print should say — and nobody regards them as fraudulent.

So 'the camera does not lie' was always false, and anyone framing the current situation as a fall from a prior state of mechanical objectivity is mistaken about the history.

But the degree has changed by orders of magnitude. A tone curve interprets a measurement; a generative fill produces content that was never in front of the lens. **A sufficiently large difference of degree is worth treating as a difference in kind,** and the appropriate response is to be precise rather than to deny that anything has changed.

Operation	What it is
Exposure, white balance, tone curve	Interpretation of a measurement. The data constrains the result tightly; the choices are about presentation (Modules 04, 05).
Demosaicing	Interpolation. Two-thirds of the colour data is produced rather than measured — and tightly constrained by the surrounding measurements (Module 05).
Burst merging	Still measurement, and more of it. Averaging aligned frames produces an estimate that converges to the truth with more frames (Module 10).

Operation	What it is
Classical denoising	Estimation under an explicit noise model. The prior is simple and stated (Module 01).
Learned denoising and upscaling	Inference from a learned prior. The result is plausible given the data and the training distribution, and it is not determined by the data (Module 12).
Generative fill, sky replacement, object removal	Synthesis. Content that was not photographed.

There is no sharp line, and the gradient is real. Attempts to legislate a boundary — 'adjustments are acceptable and manipulations are not' — fail immediately on contact with the cases. **The obligation is not to locate a line but to say where on the spectrum a given image sits**, which is a question anyone who understands their pipeline can answer.

2 Provenance

- **C2PA and Content Credentials.** Cryptographically signed metadata recording the capture device and every subsequent edit, forming a verifiable chain. Backed by camera manufacturers, software vendors, and news organisations. **The most serious current effort** and the one to know by name.
- **Signed capture.** The camera signs the raw file at the moment of capture, so any later alteration invalidates the signature. Requires hardware support and secure key storage.
- **Forensic analysis.** Detect manipulation after the fact from sensor pattern noise (every sensor has a unique fingerprint), inconsistent compression history, impossible lighting or shadow geometry, and resampling traces. Works, and is an arms race.
- **Watermarking generated content**, so that synthetic images identify themselves. Requires the generator's cooperation, which means it addresses responsible actors only.
- **Each has real and specific limits**, and the one in the callout below applies to all of them.

PROVENANCE ESTABLISHES HISTORY, NOT TRUTH

A signed chain of custody tells you what was done to an image. It does not tell you whether the image depicts what it appears to depict.

A photograph of a staged scene is entirely unmanipulated and completely misleading. So is a true photograph with a false caption, or one cropped to exclude context, or one taken at a moment unrepresentative of events. **Signing changes none of this**, and it is the larger share of how photographs actually mislead.

And the chain breaks easily. A screenshot, a re-encode, an upload through a platform that strips metadata, or a crop by a tool that does not participate — any of these severs it, and all of them are routine.

Which creates the asymmetry that limits the whole approach: absence of credentials cannot be taken to mean an image is fake, or every legitimate photograph from a non-participating camera, or any image that passed through ordinary tooling, stands condemned. **So the signal is positive-only and therefore much weaker than it first appears** — it can support a claim of authenticity and cannot refute one.

3 Evaluating claims

- 1 **What was measured, and what was inferred?** The question this course has returned to in every module. For any output, ask which parts are constrained by the data and which by a prior.
- 2 **What is the ground truth, and how was it obtained?** A comparison against another algorithm's output is not a comparison against reality. Ground truth for restoration means a genuinely clean capture — a long exposure, a controlled setup, a higher-quality instrument — and if none was available, the evaluation is relative rather than absolute.
- 3 **Which metric, and on which axis of the tradeoff?** PSNR favours blur; perceptual metrics favour plausibility; they are opposed by a theorem (Module 12). A method cannot be better on both past the bound, so a claim of being better needs to say better at what.
- 4 **What are the failure cases, and were any shown? A paper or product with no failure figure has not looked,** or has looked and declined to report. Every method in this course has inputs that defeat it, and knowing them is more useful than another percentage point.
- 5 **Is the comparison equal-effort?** A carefully tuned new method against a baseline run at its defaults is not a comparison. Nor is a comparison at equal iterations when the methods have different per-iteration costs — the equal-time discipline from CSCE 647 applies here too.

These questions generalise well beyond computational photography, and they are probably the most transferable thing in this module.

THE OBLIGATIONS ARE MODEST AND SPECIFIC

Say what you did. If an image is a fifteen-frame burst merge with a learned denoiser and a local tone map, say so — in a caption, a methods section, or the metadata. This costs nothing and it is the whole of the obligation in most contexts.

Distinguish the pleasing result from the measurement, and keep both when the measurement matters. In science, medicine, forensics, and journalism, the processed image is a derived product and the measurement should be retained and available.

Do not present synthesis as evidence. An upscaled licence plate is not a reading of a licence plate (Module 12), a reconstructed face is not an identification, and an inpainted region is not a record of what was there. This is the one that has real consequences.

And know which you are doing — which requires understanding the pipeline you are using, including the parts a vendor has not documented. **That understanding is what this course was for.**

4 Where this leaves you

- **You can take a raw file and produce a displayable image through code you wrote,** and justify every stage between the two — black level, white balance, demosaic, colour matrix, denoise, tone map, encode.

- **You can merge exposures, align and merge bursts, composite seamlessly, estimate depth, and render synthetic defocus**, and you have measured the results against ground truth you captured yourself.
- **You can recognise an underdetermined problem** — which is nearly all of them — **and identify the prior being used to resolve it**, whether it is hand-designed or learned.
- **You can read a computational imaging claim critically** and ask the five questions in §3 of it.
- **And you can say, of any image you produce, which parts were measured and which were inferred.** **That is the course's actual thesis**, and it is the thing worth retaining when the specific algorithms have been superseded — which, given the pace of this field, they will be.

THE CONNECTIONS ACROSS THE PROGRAM

CSCE 647 computed images from scene descriptions; this course recovered scene properties from images. The two share more than the subject suggests: radiance invariance underlies both ray tracing and the 4D light field (Module 09), and the rule that all processing happens in linear space with encoding last was reached independently from both directions.

CSCE 649's pressure projection, Module 08's seamless cloning, and Module 11's matting Laplacian are the same sparse linear system, arrived at three times from three unrelated problems. Recognising a Poisson problem is worth more than any individual algorithm.

CSCE 650 put a person inside the result and made the constraint physiological; **CSCE 735 makes any of it fast enough to matter.**

And the discipline has been the same throughout the program: know what your method assumes, know where it fails, measure rather than assert, and be explicit about which of your claims are measurements and which are inferences. **That is more durable than any of the techniques**, and it is what the whole degree was for.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
C2PA — Coalition for Content Provenance and Authenticity (free specification)	The provenance standard of §2. The specification is readable and the threat model section is the honest part.
Farid — Photo Forensics	The forensic analysis of §2, from the leading researcher. His papers and course material are free.
Levoy — writings on computational photography and what it means (free)	Thoughtful on the §1 question from someone who built the pipelines that raised it.
Reuters and AP photo manipulation standards (free)	How news organisations actually draw the line — worth reading because they had to be specific where academics can be vague.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Take one of your own photographs and write a complete account of every transformation between the sensor and the file, including the ones your camera performed without telling you.
- 2 Mark, on one output image from your Project 2 pipeline, the regions that are measurement, interpolation, estimation, and synthesis.
- 3 Examine the Content Credentials of an image that has them, and then screenshot it and observe what survives.
- 4 Find a photograph that is unmanipulated and misleading, and one that is manipulated and honest. Write a paragraph on each.
- 5 Take a published computational imaging result and apply the five questions of §3. Report which it answers.
- 6 Find a paper with no failure cases shown, and construct an input that defeats its method.
- 7 Write the caption you would attach to a night-mode photograph submitted as evidence, describing what it does and does not document.
- 8 Compare the photo manipulation standards of two news organisations and identify where they differ.
- 9 **Project 2 is now due.** Submit the pipeline, the quantitative evaluation against ground truth, the stage-by-stage figure, the documented failure, and the written account of what was measured and what was inferred.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why was photography never a pure measurement, and what has actually changed?
- 2 Place six operations on the spectrum from interpretation to synthesis.
- 3 Name four provenance approaches.
- 4 What does provenance establish, and what does it not?
- 5 Why is absence of credentials not evidence of fakery, and what does that imply?
- 6 Give the five questions to ask of any computational imaging claim.
- 7 State the four obligations that follow for a practitioner.
- 8 What is the course's thesis in one sentence?
- 9 Name three connections between this course and others in the program.